

Computational Companion to “Generalized Quasi-Symmetric Nets: A Constructive Approach to the Equimodular Elliptic Type of $n \times m$ Kokotsakis Polyhedra” — Helper

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Tested on: Mathematica 14.0

Section 1

Results of Section 1 are used in the proof of Lemmas 1-2 (see paper)

In[248]:=

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(* =====*)
(*PART 1:INITIALIZATION& DOMAIN ASSUMPTIONS*)
(* =====*)
ClearAll[ $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta$ ,  $\sigma$ , a, b, c, d, r, s, f,
  M, assumptions, isTrueQ, statusChip, row, identities];
 $\sigma = (\alpha + \beta + \gamma + \delta) / 2$ ;
a = Sin[ $\alpha$ ] / Sin[ $\sigma - \alpha$ ]; b = Sin[ $\beta$ ] / Sin[ $\sigma - \beta$ ];
c = Sin[ $\gamma$ ] / Sin[ $\sigma - \gamma$ ]; d = Sin[ $\delta$ ] / Sin[ $\sigma - \delta$ ];
r = a d; s = c d; f = a c; M = a b c d;
(*Domain assumptions:the flat angles lie in (0,Pi)*)
assumptions = {0 <  $\alpha$  < Pi, 0 <  $\beta$  < Pi, 0 <  $\gamma$  < Pi, 0 <  $\delta$  < Pi};
isTrueQ[lhs_, rhs_] :=
  TrueQ[FullSimplify[lhs == rhs, Assumptions  $\rightarrow$  assumptions]];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background  $\rightarrow$  RGBColor[0.13, 0.55, 0.13], FrameStyle  $\rightarrow$  None,
  RoundingRadius  $\rightarrow$  7, FrameMargins  $\rightarrow$  {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
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Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
(* =====*)
(*PART 2:THE SIX RATIO IDENTITIES*)
(* =====*)
identities = {"Identity 1", HoldForm[Sin[α] Sin[δ] / (Sin[β] Sin[γ])],
  HoldForm[r (r - M) / (M (r - 1))], isTrueQ},
{"Identity 2", HoldForm[Sin[σ - α] Sin[σ - δ] / (Sin[σ - β] Sin[σ - γ])],
  HoldForm[(r - M) / (r (r - 1))], isTrueQ},
{"Identity 3", HoldForm[Sin[γ] Sin[δ] / (Sin[α] Sin[β])],
  HoldForm[s (s - M) / (M (s - 1))], isTrueQ},
{"Identity 4", HoldForm[Sin[σ - γ] Sin[σ - δ] / (Sin[σ - α] Sin[σ - β])],
  HoldForm[(s - M) / (s (s - 1))], isTrueQ},
{"Identity 5", HoldForm[Sin[α] Sin[γ] / (Sin[β] Sin[δ])],
  HoldForm[f (f - M) / (M (f - 1))], isTrueQ},
{"Identity 6", HoldForm[Sin[σ - α] Sin[σ - γ] / (Sin[σ - β] Sin[σ - δ])],
  HoldForm[(f - M) / (f (f - 1))], isTrueQ}};
(* =====*)
(*PART 3:VERIFICATION TABLE*)
(* =====*)
row[{desc_, lhs_, rhs_, test_}] :=
  {Style[desc, 13], Item[Style[TraditionalForm[lhs], 16], Alignment → Center],
   Item[Style[TraditionalForm[rhs], 16], Alignment → Center],
   statusChip[test[ReleaseHold@lhs, ReleaseHold@rhs]]};
Grid[
  Join[{{Style["Description", Bold, 14], Style["Left Hand Side (LHS)", Bold, 14],
    Style["Right Hand Side (RHS)", Bold, 14],
    Style["Equality Status", Bold, 14]}}, row /@ identities],
  Frame → All, FrameStyle → GrayLevel[0.75],
  Background → {None, {1 → RGBColor[0.99, 0.96, 0.88]}},
  Spacings → {2.2, 2.8},
  Alignment → {{Left, Center, Center, Center}, Center},
  ItemStyle → Directive[FontFamily → "Helvetica", 13]]

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Out[259]=

Description	Left Hand Side (LHS)	Right Hand Side (RHS)	Equality Status
Identity 1	$\frac{\sin(\alpha) \sin(\delta)}{\sin(\beta) \sin(\gamma)}$	$\frac{r(r-M)}{M(r-1)}$	Verified
Identity 2	$\frac{\sin(\sigma-\alpha) \sin(\sigma-\delta)}{\sin(\sigma-\beta) \sin(\sigma-\gamma)}$	$\frac{r-M}{r(r-1)}$	Verified
Identity 3	$\frac{\sin(\gamma) \sin(\delta)}{\sin(\alpha) \sin(\beta)}$	$\frac{s(s-M)}{M(s-1)}$	Verified
Identity 4	$\frac{\sin(\sigma-\gamma) \sin(\sigma-\delta)}{\sin(\sigma-\alpha) \sin(\sigma-\beta)}$	$\frac{s-M}{s(s-1)}$	Verified
Identity 5	$\frac{\sin(\alpha) \sin(\gamma)}{\sin(\beta) \sin(\delta)}$	$\frac{f(f-M)}{M(f-1)}$	Verified
Identity 6	$\frac{\sin(\sigma-\alpha) \sin(\sigma-\gamma)}{\sin(\sigma-\beta) \sin(\sigma-\delta)}$	$\frac{f-M}{f(f-1)}$	Verified

Section 2

Results of Section 2 are used in the proof of Lemma 2 (see paper)

In[241]:=

```
( * ===== *)
(*PART 1:INITIALIZATION*)
( * ===== *)
ClearAll[f, l, m, M, isTrueQ, statusChip, row, identities];
(*The identities below are identities of
rational functions in Subscript[f,l],Subscript[f,m],M;
the subscripted symbols remain inert,so no domain assumptions are needed*)
isTrueQ[lhs_, rhs_] := TrueQ[Simplify[lhs == rhs]];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background -> RGBColor[0.13, 0.55, 0.13], FrameStyle -> None,
  RoundingRadius -> 7, FrameMargins -> {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background -> RGBColor[0.82, 0.10, 0.10], FrameStyle -> None,
  RoundingRadius -> 7, FrameMargins -> {{12, 12}, {4, 4}}];
( * ===== *)
(*PART 2:THE FOUR FACTORIZATION IDENTITIES*)
( * ===== *)
identities = {"Factorization 1",
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HoldForm[Subscript[f, l] (Subscript[f, l] - M) / (M (Subscript[f, l] - 1)) -
  Subscript[f, m] (Subscript[f, m] - M) / (M (Subscript[f, m] - 1))],
HoldForm[(Subscript[f, l] - Subscript[f, m])
  (Subscript[f, l] Subscript[f, m] - (Subscript[f, l] + Subscript[f, m]) + M) /
  (M (Subscript[f, l] - 1) (Subscript[f, m] - 1))],
isTrueQ}, {"Factorization 2", HoldForm[
  (Subscript[f, l] (Subscript[f, l] - M) / (M (Subscript[f, l] - 1))) *
  (Subscript[f, m] (Subscript[f, m] - M) / (M (Subscript[f, m] - 1))) - 1],
HoldForm[(Subscript[f, l] Subscript[f, m] - M) (Subscript[f, l]
  Subscript[f, m] - M (Subscript[f, l] + Subscript[f, m] - 1)) /
  (M^2 (Subscript[f, l] - 1) (Subscript[f, m] - 1))], isTrueQ},
{"Factorization 3", HoldForm[(Subscript[f, l] - M) /
  (Subscript[f, l] (Subscript[f, l] - 1)) -
  (Subscript[f, m] - M) / (Subscript[f, m] (Subscript[f, m] - 1))],
HoldForm[(Subscript[f, m] - Subscript[f, l]) (Subscript[f, l]
  Subscript[f, m] - M (Subscript[f, l] + Subscript[f, m] - 1)) /
  (Subscript[f, l] Subscript[f, m] (Subscript[f, l] - 1)
  (Subscript[f, m] - 1))], isTrueQ}, {"Factorization 4",
HoldForm[((Subscript[f, l] - M) / (Subscript[f, l] (Subscript[f, l] - 1))) *
  ((Subscript[f, m] - M) / (Subscript[f, m] (Subscript[f, m] - 1))) - 1],
HoldForm[(Subscript[f, l] Subscript[f, m] - M)
  (Subscript[f, l] + Subscript[f, m] - Subscript[f, l] Subscript[f, m] - M) /
  (Subscript[f, l] Subscript[f, m] (Subscript[f, l] - 1)
  (Subscript[f, m] - 1))], isTrueQ}}];

(* =====*)
(*PART 3:VERIFICATION TABLE*)
(* =====*)
row[{desc_, lhs_, rhs_, test_}] :=
  {Style[desc, 13], Item[Style[TraditionalForm[lhs], 16], Alignment → Center],
   Item[Style[TraditionalForm[rhs], 16], Alignment → Center],
   statusChip[test[ReleaseHold@lhs, ReleaseHold@rhs]]];
Grid[
  Join[{{Style["Description", Bold, 14], Style["Left Hand Side (LHS)", Bold, 14],
    Style["Right Hand Side (RHS)", Bold, 14],
    Style["Equality Status", Bold, 14]}}, row /@ identities],
  Frame → All, FrameStyle → GrayLevel[0.75],
  Background → {None, {1 → RGBColor[0.99, 0.96, 0.88]}},
  Spacings → {2.2, 2.8},
  Alignment → {{Left, Center, Center, Center}, Center},
  ItemStyle → Directive[FontFamily → "Helvetica", 13]]

```

Out[247]=

Description	Left Hand Side (LHS)	Right Hand Side (RHS)	Equality Status
Factorization 1	$\frac{f_l(f_l-M)}{M(f_l-1)} - \frac{f_m(f_m-M)}{M(f_m-1)}$	$\frac{(f_l-f_m)(f_l f_m - (f_l+f_m)+M)}{M(f_l-1)(f_m-1)}$	Verified
Factorization 2	$\frac{(f_l(f_l-M))(f_m(f_m-M))}{(M(f_l-1))(M(f_m-1))} - 1$	$\frac{(f_l f_m - M)(f_l f_m - M(f_l+f_m-1))}{M^2(f_l-1)(f_m-1)}$	Verified
Factorization 3	$\frac{f_l-M}{f_l(f_l-1)} - \frac{f_m-M}{f_m(f_m-1)}$	$\frac{(f_m-f_l)(f_l f_m - M(f_l+f_m-1))}{f_l f_m(f_l-1)(f_m-1)}$	Verified
Factorization 4	$\frac{(f_l-M)(f_m-M)}{(f_l(f_l-1))(f_m(f_m-1))} - 1$	$\frac{(f_l f_m - M)(f_l+f_m-f_l f_m - M)}{f_l f_m(f_l-1)(f_m-1)}$	Verified

Section 3

Results of Section 3 are used in the proof of Lemma 4 (see paper)

In[226]:=

```
(* =====*)
(*PART 1:INITIALIZATION& DOMAIN ASSUMPTIONS*)
(* =====*)
ClearAll[α, β, γ, δ, σ, ε, a, b, c, d, r, s, f, M, rad, sq,
  assumptions, isTrueQ, isTrueQeps, statusChip, row, identities];
σ = (α + β + γ + δ) / 2;
a = Sin[α] / Sin[σ - α]; b = Sin[β] / Sin[σ - β];
c = Sin[γ] / Sin[σ - γ]; d = Sin[δ] / Sin[σ - δ];
r = a d; s = c d; f = a c; M = a b c d;
(*Domain
  assumptions:the flat angles and the sigma-shifted angles lie in (0,Pi);
these hold under the hypotheses of the lemma
  indicated in the subsection above (see paper)*)
assumptions = {0 < α < Pi, 0 < β < Pi, 0 < γ < Pi,
  0 < δ < Pi, 0 < σ - α < Pi, 0 < σ - β < Pi, 0 < σ - γ < Pi, 0 < σ - δ < Pi};
rad = (f - M) (r - 1) (s - 1) / (f (1 - M));
sq = (Sin[σ] / Sin[σ - δ]) ^ 2;
isTrueQ[lhs_, rhs_] :=
  TrueQ[FullSimplify[lhs == rhs, Assumptions → assumptions]];
(*Two-stage verification,
in the order of the table:STAGE 1 (the radicand row) identifies rad with the
  perfect square sq.STAGE 2 (the identity row) uses Stage 1:for t=
  Sin[sigma]/Sin[sigma-delta],the eps-split assumptions
  give t the sign eps (u/v checks),
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so Sqrt[rad]=Sqrt[t^2]=eps*t (abstract root law),
and eps*Sqrt[rad]*Sin[sigma-delta]=
  eps^2*Sin[sigma]=Sin[sigma] (cancellation).No Sqrt
  over trigonometric arguments reaches the kernel.*)
isTrueQeps[lhs_, rhs_] :=
  Module[{t, u, v}, TrueQ[FullSimplify[rad == sq, Assumptions → assumptions]] &&
    TrueQ[Assuming[u > 0 && v > 0, Simplify[u / v > 0]]] &&
    TrueQ[Assuming[u < 0 && v > 0, Simplify[u / v < 0]]] &&
    TrueQ[Assuming[t > 0, Simplify[Sqrt[t^2] == t]]] &&
    TrueQ[Assuming[t < 0, Simplify[Sqrt[t^2] == -t]]] &&
    TrueQ[Simplify[(Sin[σ] / Sin[σ - δ]) Sin[σ - δ] == Sin[σ]]];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
(* =====*)
(*PART 2:THE RADICAND AND THE IDENTITY*)
(* =====*)
identities = {"Radicand equals a perfect square",
  HoldForm[(f - M) (r - 1) (s - 1) / (f (1 - M))], HoldForm[(Sin[σ] / Sin[σ - δ]) ^ 2],
  isTrueQ}, {"Identity for sin(σ) (via the radicand row)", HoldForm[Sin[σ]],
  HoldForm[ε Sqrt[(f - M) (r - 1) (s - 1) / (f (1 - M))] Sin[σ - δ]], isTrueQeps};
(* =====*)
(*PART 3:VERIFICATION TABLE*)
(* =====*)
row[{desc_, lhs_, rhs_, test_}] :=
  {Style[desc, 13], Item[Style[TraditionalForm[lhs], 16], Alignment → Center],
  Item[Style[TraditionalForm[rhs], 16], Alignment → Center],
  statusChip[test[ReleaseHold@lhs, ReleaseHold@rhs]]};
Grid[
  Join[{{Style["Description", Bold, 14], Style["Left Hand Side (LHS)", Bold, 14],
    Style["Right Hand Side (RHS)", Bold, 14],
    Style["Equality Status", Bold, 14]}}, row /@ identities],
  Frame → All, FrameStyle → GrayLevel[0.75],
  Background → {None, {1 → RGBColor[0.99, 0.96, 0.88]}},
  Spacings → {2.2, 2.8},
  Alignment → {{Left, Center, Center, Center}, Center},
  ItemStyle → Directive[FontFamily → "Helvetica", 13]]

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Out[240]=

Description	Left Hand Side (LHS)	Right Hand Side (RHS)	Equality Status
Radicaud equals a perfect square	$\frac{(f-M)(r-1)(s-1)}{f(1-M)}$	$\left(\frac{\sin(\sigma)}{\sin(\sigma-\delta)}\right)^2$	Verified
Identity for $\sin(\sigma)$ (via the radicaud row)	$\sin(\sigma)$	$\varepsilon \sqrt{\frac{(f-M)(r-1)(s-1)}{f(1-M)}} \sin(\sigma-\delta)$	Verified

Section 4

Results of Section 4 are used in the proof of Theorem 1 (see paper)

In[281]:=

```
(* =====*)
(*PART 1:INITIALIZATION& STANDING HYPOTHESES*)
(* =====*)
ClearAll[f, i, j, l, m, c, s, si, sl, fi, fj, fl, fm,
  M, chi, x, display, propEq, standingOf, assumptions, isTrueQ,
  statusChip, caseCheck, results, starRef, fj3, fm3, fj4, fm4];
Format[fi, TraditionalForm] = Subscript[f, i];
Format[fl, TraditionalForm] = Subscript[f, l];
Format[chi, TraditionalForm] = x;
(*Standing hypotheses:M and all four f's are positive,
distinct from 1 and from each other's modulus pairing f≠M,
and each pair-product (f-1)(f-M) is positive;
these hold under the hypotheses indicated in the paper.The verification below
is branch-wise:sign data entering through chi and the regime of M is
enumerated explicitly,so every square root reaching the kernel has a sign-
determined radicaud.All display-only rows are wrapped in HoldForm,
so they cannot evaluate through symbols defined in other sections.*)
standingOf[fjE_, fmE_] := M > 0 && M ≠ 1 && And@@ (# > 0 & /@ {fi, fjE, fl, fmE}) &&
  And@@ (# ≠ 1 & /@ {fi, fjE, fl, fmE}) && And@@ (# ≠ M & /@ {fi, fjE, fl, fmE}) &&
  And@@ ((# - 1) (# - M) > 0 & /@ {fi, fjE, fl, fmE});
isTrueQ[lhs_, rhs_] :=
  TrueQ[FullSimplify[lhs == rhs, Assumptions → assumptions]];
display[zeta1_, zeta2_, chiv_, fjE_, fmE_] :=
  zeta1 chiv M Sqrt[(fi - 1) (fjE - 1) (fl - 1) (fmE - 1)] + zeta2 (1 - M)
  Sqrt[fi fjE fl fmE] - zeta2 chiv Sqrt[(fi - M) (fjE - M) (fl - M) (fmE - M)];
propEq[zeta1_, zeta2_, chiv_, fjE_, fmE_] := Style[TraditionalForm[
  HoldForm[#] && (display[zeta1, zeta2, chiv, fjE, fmE] == M (1 - M))], 16];
starRef = Style["(*)", Bold, 16, RGBColor[0.72, 0.45, 0.02]];
```

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statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
(*Two-stage verification per
  case:STAGE 1 (the collapse row) identifies the instantiated left-hand side
    of (star) with its square-root-free form under the branch assumptions;
  STAGE 2 (the closure row) verifies that the collapsed form equals
    M(1-M).The chip is Verified only if both stages hold.*/)
caseCheck[title_, col_, condRow_, zeta1_, zeta2_, branchRow_, chiv_,
  fjE_, fmE_, extraAssum_, extraRow_, collapsedHeld_] := Module[
  {okCollapse, okZero, ok}, assumptions = standingOf[fjE, fmE] && extraAssum;
  okCollapse =
    isTrueQ[display[zeta1, zeta2, chiv, fjE, fmE], ReleaseHold[collapsedHeld]];
  okZero = isTrueQ[ReleaseHold[collapsedHeld], M (1 - M)];
  ok = okCollapse && okZero;
  Print[Panel[Column[
    {Row[{Style[title, Bold, 15, col], Spacer[16], Style[condRow, 13, col]}],
      Row[{Style["testing ", 13], starRef,
        Style[Row[{" with ", Subscript["g", 1], " = ", zeta1, " ",
          Subscript["g", 2], " = ", zeta2}], Bold, 13], Spacer[24],
        branchRow}], propEq[zeta1, zeta2, chiv, fjE, fmE], Row[
        {Style["expanding the square roots: ", GrayLevel[0.35], 12], Style[
          TraditionalForm[HoldForm[#] &@ (collapsedHeld == M (1 - M))], 13}],
        Row[{Style["→ ", Bold, 14], Style[TraditionalForm[HoldForm[0 == 0]],
          14], Spacer[26], statusChip[ok]}],
        Style[Row[{"additionally used: ", extraRow}], GrayLevel[0.35], 12}],
        Spacings → 1], Background → Lighter[col, 0.93], FrameMargins → 12]];
  ok];
(* =====*)
(*PART 2:STANDING HYPOTHESES AND THE IDENTITY (star)*)
(* =====*)
Print[Panel[Column[{Style["Standing Hypotheses:", Bold, 14],
  Style[Row[{TraditionalForm[HoldForm[Subscript[f, i]], " ",
    TraditionalForm[HoldForm[M]], " > 0, ", TraditionalForm[HoldForm[
      Subscript[f, i]], " ", TraditionalForm[HoldForm[M]], " ≠ 1, ",
      TraditionalForm[HoldForm[Subscript[f, i] ≠ M]], " ", TraditionalForm[
        HoldForm[(Subscript[f, i] - 1) (Subscript[f, i] - M) > 0]],
        " for each ", TraditionalForm[HoldForm[i]], " = 1, ..., 4, ",
        TraditionalForm[HoldForm[χ ∈ {+1, -1}]]}], 13], ""], Style[
    Row[{"The identity we want to verify for all cases below (for some ",
      Subscript["g", 1], " ", Subscript["g", 2], " ∈ {+1, -1}):"}, Bold,
      14], Row[{Style[TraditionalForm[HoldForm[Subscript[g, 1] χ M
        Sqrt[Product[Subscript[f, i] - 1, {i, 1, 4}]] + Subscript[g, 2]
        (1 - M) Sqrt[Product[Subscript[f, i], {i, 1, 4}]] - Subscript[g, 2]

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       $\chi \sqrt{\text{Product}[\text{Subscript}[f, i] - M, \{i, 1, 4\}]} = M (1 - M)]], 16],$ 
      Spacer[28], Style["(*)", Bold, 16, RGBColor[0.72, 0.45, 0.02]]]],
      Spacings → 1.1], Background → RGBColor[0.99, 0.96, 0.88],
      FrameMargins → 14]]];

(* =====*)
(*PART 3:CASE-BY-CASE VERIFICATION*)
(* =====*)
results = {};
(*Condition (i):fj=fi, fm=fl;
chi (fi-1) (fl-1)>0;
zeta1=zeta2=1.The four branches enumerate the signs of fi-1 and fl-1;
by the standing pair-products,fi-M and fi-1 have the same sign,
and similarly at l,so chi=si*sl.*)
Do[AppendTo[results, caseCheck["Condition (i):", RGBColor[0.10, 0.30, 0.70],
  Row[{TraditionalForm[HoldForm[Subscript[f, j] == Subscript[f, i]]],
    ", ", TraditionalForm[HoldForm[Subscript[f, m] == Subscript[f, l]]],
    ", ", TraditionalForm[
      HoldForm[ $\chi (\text{Subscript}[f, i] - 1) (\text{Subscript}[f, l] - 1) > 0$ ]]], 1,
    1, Style[Row[{"branch: sign(", TraditionalForm[HoldForm[fi - 1]],
      ") = ", si, ", sign(", TraditionalForm[HoldForm[fl - 1]],
      ") = ", sl, " →  $\chi =$ ", si sl}], 13, GrayLevel[0.3]], si sl, fi,
    fl, si (fi - 1) > 0 && si (fi - M) > 0 && sl (fl - 1) > 0 && sl (fl - M) > 0,
    Row[{TraditionalForm[HoldForm[#] && (si (fi - 1) > 0)],
      ", ", TraditionalForm[HoldForm[#] && (si (fi - M) > 0)],
      ", ", TraditionalForm[HoldForm[#] && (sl (fl - 1) > 0)],
      ", ", TraditionalForm[HoldForm[#] && (sl (fl - M) > 0)],
      " (same-sign facts by the standing hypotheses)"}],
    HoldForm[M (fi - 1) (fl - 1) + (1 - M) fi fl - (fi - M) (fl - M)]],
    {si, {1, -1}}, {sl, {1, -1}}];
(*Condition (ii):fi*fj=M, fl*fm=M, that is, fj=M/fi, fm=M/fl;
zeta1=zeta2=1;
chi stays symbolic (no sign condition is consumed).*)
AppendTo[results, caseCheck["Condition (ii):", RGBColor[0.55, 0.25, 0.70],
  Row[{TraditionalForm[HoldForm[Subscript[f, i] Subscript[f, j] == M]], ", ",
    TraditionalForm[HoldForm[Subscript[f, l] Subscript[f, m] == M]]], 1,
    1, Style["single symbolic branch", 13, GrayLevel[0.3]], chi, M / fi,
    M / fl, True, "none ( $\chi$  remains symbolic; no sign condition is consumed)",
    HoldForm[chi M Sqrt[(fi - 1) (M / fi - 1) (fl - 1) (M / fl - 1)] +
      (1 - M) M - chi M Sqrt[(fi - 1) (M / fi - 1) (fl - 1) (M / fl - 1)]]];
(*Condition (iii):fi+fj-fi*fj=M, fl+fm-fl*fm=M,
that is, fj=(M-fi)/(1-fi), fm=(M-fl)/(1-fl);
chi (1-M)>0;
zeta1=zeta2=1.Two branches:chi=+1 with M<1,and chi=-1 with M>1.*)
fj3 = (M - fi) / (1 - fi); fm3 = (M - fl) / (1 - fl);
Do[AppendTo[results, caseCheck["Condition (iii):",
  RGBColor[0.00, 0.50, 0.50], Row[{TraditionalForm[HoldForm[
    Subscript[f, i] + Subscript[f, j] - Subscript[f, i] Subscript[f, j] == M]],

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", TraditionalForm[HoldForm[
  Subscript[f, l] + Subscript[f, m] - Subscript[f, l] Subscript[f, m] == M]],
", TraditionalForm[HoldForm[ $\chi (1 - M) > 0$ ]]], 1, 1,
Style[Row[{"branch:  $\chi =$  ", c, " ", " ", If[c == 1, "M < 1", "M > 1"]}],
13, GrayLevel[0.3]], c, fj3, fm3, If[c == 1, M < 1, M > 1],
Row[{TraditionalForm[HoldForm[#] &@If[c == 1, M < 1, M > 1]],
" (this branch of ",
TraditionalForm[HoldForm[ $\chi (1 - M) > 0$ ]], ")"], HoldForm[
M (1 - M) + (1 - M) Sqrt[fi ((M - fi) / (1 - fi)) fl ((M - fl) / (1 - fl))] - (1 - M)
Sqrt[fi ((M - fi) / (1 - fi)) fl ((M - fl) / (1 - fl))]]], {c, {1, -1}}];
(*Condition (iv):M(fi+fj-1)=fi*fj,M(fl+fm-1)=fl*fm,
that is,fj=M(fi-1)/(fi-M),fm=M(fl-1)/(fl-M);
zeta1=1,and zeta2=-1 if chi(1-M)>0,zeta2=+1 if chi(1-M)<0,encoded as zeta2=
-c*s with s=sign(1-M).Four branches over chi and the regime of M.*)
fj4 = M (fi - 1) / (fi - M); fm4 = M (fl - 1) / (fl - M);
Do[AppendTo[results, caseCheck["Condition (iv):", RGBColor[0.75, 0.35, 0.05],
Row[{TraditionalForm[HoldForm[M (Subscript[f, i] + Subscript[f, j] - 1) ==
Subscript[f, i] Subscript[f, j]]], " ", " ",
TraditionalForm[HoldForm[M (Subscript[f, l] + Subscript[f, m] - 1) ==
Subscript[f, l] Subscript[f, m]]]], 1, -c s,
Style[Row[{"branch:  $\chi =$  ", c, " ", " ", If[s == 1, "M < 1", "M > 1"],
"  $\rightarrow$  ", If[c s == 1, " $\chi(1-M) > 0$ ", " $\chi(1-M) < 0$ "]}],
13, GrayLevel[0.3]], c, fj4, fm4, If[s == 1, M < 1, M > 1],
Row[{TraditionalForm[HoldForm[#] &@If[s == 1, M < 1, M > 1]],
" (this branch of the rule: ", Subscript[" $\xi$ ", 2],
" = -1 if ", TraditionalForm[HoldForm[ $\chi (1 - M) > 0$ ]],
", " ", Subscript[" $\xi$ ", 2], " = +1 if ",
TraditionalForm[HoldForm[ $\chi (1 - M) < 0$ ]], ")"], HoldForm[#] &@
(c s (1 - M) Sqrt[fi (M (fi - 1) / (fi - M)) fl (M (fl - 1) / (fl - M))] -
c s (1 - M) Sqrt[fi (M (fi - 1) / (fi - M)) fl (M (fl - 1) / (fl - M))] +
M (1 - M))]], {c, {1, -1}}, {s, {1, -1}}];
(* =====*)
(*PART 4:SUMMARY*)
(* =====*)
Print[
Framed[Style[If[And@@results, "ALL CASES VERIFIED", "SOME CASES FAILED"],
Bold, White, 16], Background  $\rightarrow$ 
If[And@@results, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
FrameStyle  $\rightarrow$  None, RoundingRadius  $\rightarrow$  10, FrameMargins  $\rightarrow$  {{18, 18}, {8, 8}}]];

```

Standing Hypotheses:

$f_i, M > 0, f_i, M \neq 1, f_i \neq M, (f_i - 1)(f_i - M) > 0$ for each $i = 1, \dots, 4, \chi \in \{+1, -1\}$

The identity we want to verify for all cases below (for some $\zeta_1, \zeta_2 \in \{+1, -1\}$):

$$\zeta_1 \chi M \sqrt{\prod_{i=1}^4 (f_i - 1)} + \zeta_2 (1 - M) \sqrt{\prod_{i=1}^4 f_i} - \zeta_2 \chi \sqrt{\prod_{i=1}^4 (f_i - M)} = M(1 - M)$$

(★)

Condition (i): $f_j = f_i, f_m = f_l, \chi (f_i - 1)(f_l - 1) > 0$

testing (★) with $\zeta_1 = 1, \zeta_2 = 1$ branch: $\text{sign}(f_i - 1) = 1, \text{sign}(f_l - 1) = 1 \rightarrow \chi = 1$

$$\sqrt{f_i^2 f_l^2} (1 - M) - \sqrt{(f_i - M)^2 (f_l - M)^2} + \sqrt{(-1 + f_i)^2 (-1 + f_l)^2} M = (1 - M) M$$

expanding the square roots: $M(f_i - 1)(f_l - 1) + (1 - M)f_i f_l - (f_i - M)(f_l - M) = (1 - M)M$

$\rightarrow 0 = 0$

Verified

additionally used:

$-1 + f_i > 0, f_i - M > 0, -1 + f_l > 0, f_l - M > 0$ (same-sign facts by the standing hypotheses)

Condition (i): $f_j = f_i, f_m = f_l, \chi (f_i - 1)(f_l - 1) > 0$

testing (★) with $\zeta_1 = 1, \zeta_2 = 1$ branch: $\text{sign}(f_i - 1) = 1, \text{sign}(f_l - 1) = -1 \rightarrow \chi = -1$

$$\sqrt{f_i^2 f_l^2} (1 - M) + \sqrt{(f_i - M)^2 (f_l - M)^2} - \sqrt{(-1 + f_i)^2 (-1 + f_l)^2} M = (1 - M) M$$

expanding the square roots: $M(f_i - 1)(f_l - 1) + (1 - M)f_i f_l - (f_i - M)(f_l - M) = (1 - M)M$

$\rightarrow 0 = 0$

Verified

additionally used:

$-1 + f_i > 0, f_i - M > 0, 1 - f_l > 0, -f_l + M > 0$ (same-sign facts by the standing hypotheses)

Condition (i): $f_j = f_i, f_m = f_l, \chi (f_i - 1)(f_l - 1) > 0$

testing (★) with $\zeta_1 = 1, \zeta_2 = 1$ branch: $\text{sign}(f_i - 1) = -1, \text{sign}(f_l - 1) = 1 \rightarrow \chi = -1$

$$\sqrt{f_i^2 f_l^2} (1 - M) + \sqrt{(f_i - M)^2 (f_l - M)^2} - \sqrt{(-1 + f_i)^2 (-1 + f_l)^2} M = (1 - M) M$$

expanding the square roots: $M(f_i - 1)(f_l - 1) + (1 - M)f_i f_l - (f_i - M)(f_l - M) = (1 - M)M$

$\rightarrow 0 = 0$

Verified

additionally used:

$1 - f_i > 0, -f_i + M > 0, -1 + f_l > 0, f_l - M > 0$ (same-sign facts by the standing hypotheses)

Condition (i): $f_j = f_i, f_m = f_l, \chi(f_i - 1)(f_l - 1) > 0$

testing (★) with $\zeta_1 = 1, \zeta_2 = 1$ branch: $\text{sign}(f_i - 1) = -1, \text{sign}(f_l - 1) = -1 \rightarrow \chi = 1$

$$\sqrt{f_i^2 f_l^2} (1 - M) - \sqrt{(f_i - M)^2 (f_l - M)^2} + \sqrt{(-1 + f_i)^2 (-1 + f_l)^2} M = (1 - M) M$$

expanding the square roots: $M(f_i - 1)(f_l - 1) + (1 - M)f_i f_l - (f_i - M)(f_l - M) = (1 - M) M$

$$\rightarrow 0 = 0$$

Verified

additionally used:

$1 - f_i > 0, -f_i + M > 0, 1 - f_l > 0, -f_l + M > 0$ (same-sign facts by the standing hypotheses)

Condition (ii): $f_i f_j = M, f_l f_m = M$

testing (★) with $\zeta_1 = 1, \zeta_2 = 1$ single symbolic branch

$$(1 - M) \sqrt{M^2} + \chi M \sqrt{(-1 + f_i)(-1 + f_l) \left(-1 + \frac{M}{f_i}\right) \left(-1 + \frac{M}{f_l}\right)} -$$

$$\chi \sqrt{(f_i - M)(f_l - M) \left(-M + \frac{M}{f_i}\right) \left(-M + \frac{M}{f_l}\right)} = (1 - M) M$$

expanding the square roots: $\chi M \sqrt{(f_i - 1) \left(\frac{M}{f_i} - 1\right) (f_l - 1) \left(\frac{M}{f_l} - 1\right)} +$

$$(1 - M) M - \chi M \sqrt{(f_i - 1) \left(\frac{M}{f_i} - 1\right) (f_l - 1) \left(\frac{M}{f_l} - 1\right)} = (1 - M) M$$

$$\rightarrow 0 = 0$$

Verified

additionally used: none (χ remains symbolic; no sign condition is consumed)

Condition (iii): $f_i + f_j - f_i f_j = M, f_l + f_m - f_l f_m = M, \chi(1 - M) > 0$

testing (★) with $\zeta_1 = 1, \zeta_2 = 1$ branch: $\chi = 1, M < 1$

$$(1 - M) \sqrt{\frac{f_i f_l (-f_i + M)(-f_l + M)}{(1 - f_i)(1 - f_l)}} + M \sqrt{(-1 + f_i)(-1 + f_l) \left(-1 + \frac{-f_i + M}{1 - f_i}\right) \left(-1 + \frac{-f_l + M}{1 - f_l}\right)} -$$

$$\sqrt{(f_i - M)(f_l - M) \left(-M + \frac{-f_i + M}{1 - f_i}\right) \left(-M + \frac{-f_l + M}{1 - f_l}\right)} = (1 - M) M$$

expanding the square roots:

$$M(1 - M) + (1 - M) \sqrt{\frac{f_i(M - f_i)f_l(M - f_l)}{(1 - f_i)(1 - f_l)}} - (1 - M) \sqrt{\frac{f_i(M - f_i)f_l(M - f_l)}{(1 - f_i)(1 - f_l)}} = (1 - M) M$$

$$\rightarrow 0 = 0$$

Verified

additionally used: $M < 1$ (this branch of $\chi(1 - M) > 0$)

Condition (iii): $f_i + f_j - f_i f_j = M$, $f_l + f_m - f_l f_m = M$, $\chi(1-M) > 0$

testing (★) with $\zeta_1 = 1$, $\zeta_2 = 1$ branch: $\chi = -1$, $M > 1$

$$(1-M) \sqrt{\frac{f_i f_l (-f_i + M) (-f_l + M)}{(1-f_i) (1-f_l)}} - M \sqrt{\left(-1 + f_i\right) \left(-1 + f_l\right) \left(-1 + \frac{-f_i + M}{1-f_i}\right) \left(-1 + \frac{-f_l + M}{1-f_l}\right)} + \\ \sqrt{(f_i - M) (f_l - M) \left(-M + \frac{-f_i + M}{1-f_i}\right) \left(-M + \frac{-f_l + M}{1-f_l}\right)} = (1-M) M$$

expanding the square roots:

$$M(1-M) + (1-M) \sqrt{\frac{f_i (M-f_i) f_l (M-f_l)}{(1-f_i) (1-f_l)}} - (1-M) \sqrt{\frac{f_i (M-f_i) f_l (M-f_l)}{(1-f_i) (1-f_l)}} = (1-M) M$$

→ 0 = 0

Verified

additionally used: $M > 1$ (this branch of $\chi(1-M) > 0$)

Condition (iv): $M(f_i + f_j - 1) = f_i f_j$, $M(f_l + f_m - 1) = f_l f_m$

testing (★) with $\zeta_1 = 1$, $\zeta_2 = -1$ branch: $\chi = 1$, $M < 1 \rightarrow \chi(1-M) > 0$

$$-\left((1-M) \sqrt{\frac{(-1+f_i) f_i (-1+f_l) f_l M^2}{(f_i-M) (f_l-M)}}\right) + \\ M \sqrt{\left(-1 + f_i\right) \left(-1 + f_l\right) \left(-1 + \frac{(-1+f_i) M}{f_i-M}\right) \left(-1 + \frac{(-1+f_l) M}{f_l-M}\right)} + \\ \sqrt{(f_i - M) (f_l - M) \left(-M + \frac{(-1+f_i) M}{f_i-M}\right) \left(-M + \frac{(-1+f_l) M}{f_l-M}\right)} = (1-M) M$$

expanding the square roots: $(1-M) M = (1-M) M$

→ 0 = 0

Verified

additionally used: $M < 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

Condition (iv): $M(f_i + f_j - 1) = f_i f_j$, $M(f_l + f_m - 1) = f_l f_m$

testing (★) with $\zeta_1 = 1$, $\zeta_2 = 1$ branch: $\chi = 1$, $M > 1 \rightarrow \chi(1-M) < 0$

$$(1-M) \sqrt{\frac{(-1+f_i) f_i (-1+f_l) f_l M^2}{(f_i-M) (f_l-M)}} + \\ M \sqrt{\left(-1 + f_i\right) \left(-1 + f_l\right) \left(-1 + \frac{(-1+f_i) M}{f_i-M}\right) \left(-1 + \frac{(-1+f_l) M}{f_l-M}\right)} - \\ \sqrt{(f_i - M) (f_l - M) \left(-M + \frac{(-1+f_i) M}{f_i-M}\right) \left(-M + \frac{(-1+f_l) M}{f_l-M}\right)} = (1-M) M$$

expanding the square roots: $(1-M) M = (1-M) M$

→ 0 = 0

Verified

additionally used: $M > 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

Condition (iv): $M(f_i + f_j - 1) = f_i f_j$, $M(f_l + f_m - 1) = f_l f_m$

testing (★) with $\zeta_1 = 1$, $\zeta_2 = 1$ branch: $\chi = -1$, $M < 1 \rightarrow \chi(1-M) < 0$

$$\begin{aligned} & (1-M) \sqrt{\frac{(-1+f_i) f_i (-1+f_l) f_l M^2}{(f_i-M)(f_l-M)}} - \\ & M \sqrt{(-1+f_i)(-1+f_l) \left(-1 + \frac{(-1+f_i)M}{f_i-M}\right) \left(-1 + \frac{(-1+f_l)M}{f_l-M}\right)} + \\ & \sqrt{(f_i-M)(f_l-M) \left(-M + \frac{(-1+f_i)M}{f_i-M}\right) \left(-M + \frac{(-1+f_l)M}{f_l-M}\right)} = (1-M)M \end{aligned}$$

expanding the square roots: $(1-M)M = (1-M)M$

$\rightarrow 0 = 0$

Verified

additionally used: $M < 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

Condition (iv): $M(f_i + f_j - 1) = f_i f_j$, $M(f_l + f_m - 1) = f_l f_m$

testing (★) with $\zeta_1 = 1$, $\zeta_2 = -1$ branch: $\chi = -1$, $M > 1 \rightarrow \chi(1-M) > 0$

$$\begin{aligned} & - \left((1-M) \sqrt{\frac{(-1+f_i) f_i (-1+f_l) f_l M^2}{(f_i-M)(f_l-M)}} \right) - \\ & M \sqrt{(-1+f_i)(-1+f_l) \left(-1 + \frac{(-1+f_i)M}{f_i-M}\right) \left(-1 + \frac{(-1+f_l)M}{f_l-M}\right)} - \\ & \sqrt{(f_i-M)(f_l-M) \left(-M + \frac{(-1+f_i)M}{f_i-M}\right) \left(-M + \frac{(-1+f_l)M}{f_l-M}\right)} = (1-M)M \end{aligned}$$

expanding the square roots: $(1-M)M = (1-M)M$

$\rightarrow 0 = 0$

Verified

additionally used: $M > 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

ALL CASES VERIFIED